
How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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Paper under double-blind review

Abstract

Spatial memory in primates is not a sensor-independent module. Hippocampal cells code joint place–view–action state in part because the primate eye samples the world through a foveated sensor; rodents with near-panoramic vision instead develop classic place fields, and individuals deprived of vision recruit hippocampal circuits to compensate. The biological pattern is consistent: *the structure of the sensor shapes the structure of spatial memory*. Whether the same holds for artificial navigation agents — whose learned memory also develops cognitive-map-like representations under task pressure alone — has not been tested directly, because prior work has either removed vision entirely or used spatially uniform input. We isolate the principle in a controlled silicon case study: five recurrent (LSTM) PointNav agents trained identically except for the visual pathway, spanning a sensor-structure spectrum from no vision, through a resolution-collapsed encoder, foveated vision (fixed and learned gaze), to full uniform vision. Probing their recurrent state reveals an *encoder–memory race*: the linearly-decodable spatial code is maintained when the encoder supplies little spatial-feature variety per step, and dissipates when the encoder is rich enough to substitute. Independently, agents at comparable task competence develop hidden states in linearly disjoint subspaces, and cross-condition memory transplants are asymmetric: bottleneck-donor states hurt rich-encoder recipients more than vice versa. The intuition that “richer perception begets richer memory” fails in this regime — the relationship across conditions is inverse — making encoder capacity a candidate training-time lever for cognitive-map-like representations (single-seed; multi-seed replication and a causal scaling sweep are in flight). The pattern echoes biological findings on hippocampal compensation and cross-modality remapping, suggesting a sensor–memory coupling principle whose generality across architectures is open.

1 Introduction

Primate spatial memory is shaped by how primates see. Hippocampal cells code joint place–view–action state in part because the primate eye samples the scene through a foveated sensor [Wirth et al., 2017, Rolls, 2024]; rodents with near-panoramic vision instead show classic place fields. Sensor structure also reorganises memory *within* a single animal — bats produce non-overlapping hippocampal populations under vision vs. echolocation [Geva-Sagiv et al., 2016], and congenitally blind humans develop enhanced hippocampal coupling on non-visual spatial tasks [Kupers and Ptito, 2014]. The biological pattern is consistent: *the structure of the sensor reshapes the structure of spatial memory*. We ask whether the same is true of artificial navigation agents.

The emergent-cognitive-maps literature has shown that map-like structure arises in the learned internal state of navigation agents trained with deep RL, including agents with no vision at all [Wijmans et al.,

2023, Banino et al., 2018, Cueva and Wei, 2018].¹ These results establish that *some* map-like structure is learned without being designed in. They leave open the complementary question we take up here: holding task and architecture fixed, does the structure of the visual sensor reshape *what* the agent’s memory encodes? We isolate sensor structure as the only varied component across five conditions spanning a sensor-structure spectrum (Figure 1): no vision, a resolution-collapsed encoder, foveated vision with fixed gaze, foveated vision with learned gaze, and uniform full-resolution vision. The resolution-collapsed control distinguishes outright blindness from the case where vision is present but the encoder cannot spatially resolve it; foveation occupies a biologically motivated point in between. *We use a recurrent (LSTM-based) PointNav agent as our case study*: this matches the architecture of the emergent-maps literature we extend [Wijmans et al., 2023], gives every cross-condition comparison a single hidden-state object to probe, and provides a tractable controlled setting. The principle we test — whether sensor structure shapes the format of learned memory — is stated in architecture-agnostic terms; whether it transfers to transformer-based or supervised visual learners is a scope question we discuss in §5, not establish here.

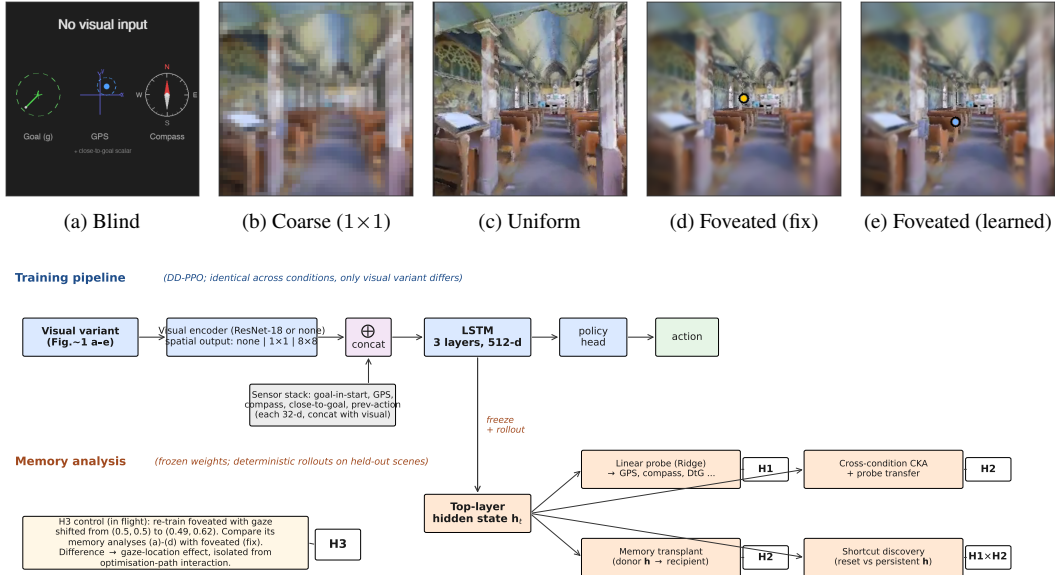


Figure 1: **Setup.** *Top row, (a)–(e):* the five visual sensor conditions, shown on the same scene. (a) *Blind*: no visual stream; only the non-visual sensor stack. (b) *Coarse* (1×1): 48×48 uniform RGB, chosen so the ResNet-18 encoder collapses spatially to a 1×1 feature map. (c) *Uniform*: 256×256 full-resolution RGB. (d) *Foveated (fix)*: eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) centred at the image middle (yellow dot). (e) *Foveated (learned)*: same blur model with a learned (x, y) gaze offset (blue dot, post-training collapsed location (0.49, 0.62)). *Bottom block:* the structural variable being manipulated is encoder spatial output (none | 1×1 | 8×8); the DD-PPO learner, sensor stack, LSTM backbone, and policy head are otherwise identical. After training, weights are frozen and we collect deterministic rollouts to extract top-layer hidden states h_t , on which we run linear probes (H1 evidence), cross-condition CKA + probe transfer (H2 evidence), memory transplants (H2 behavioural validation), and shortcut discovery (probe-readable vs. policy-used dissociation, §4.5).

We test three hypotheses. *H1 (encoder–memory race)*: when the visual encoder supplies little spatial-feature variety per step, the recurrent state should compensate by carrying a world-frame spatial code; richer encoder output should let memory rely on visual features instead. *H2 (format divergence)*: different sensor conditions should produce hidden-state representations in distinct linear subspaces, not scaled versions of a common code. *H3 (gaze location)*: within the rich-encoder regime, where a

¹We use “cognitive map” loosely throughout for a linearly-decodable world-frame position code in the agent’s hidden state — the operational quantity our probes measure. This is narrower than the cognitive-map concept in the broader neural-and-cognitive-science literature (allocentric, relational, multi-scale, hierarchical maps; see e.g. [Epstein et al., 2017]). Our claims should be read as scoped to the linearly-decodable subset; richer non-linear or relational structure may exist in the same hidden states and is not addressed by linear probes.

foveated sensor is centred should act as a second content axis independent of the sensor transform. Each has a direct biological analog (§5.2).

Three findings emerge with the strongest evidence. An encoder–memory race along the sensor-structure spectrum (H1) that lives inside the recurrent state, not at the encoder. The five agents solve the task at comparable competence but develop hidden states in linearly disjoint subspaces, with measurable cross-condition memory-transplant cost (H2). And probe-readable vs. policy-used memory dissociates in opposite directions for two specific conditions, suggesting that “has a cognitive map” and “uses a cognitive map” need not coincide. Foveation under our blur model converges with uniform on H1 but diverges on H2, behaviour, and dataset shift. H3 (gaze location) is tested by a foveated-shifted control in training at submission.

The cross-condition pattern is correlation-level; a scaling-sweep experiment that would test the H1 mechanism causally is in progress (Appendix H). Foveation-specific predictions involving alternative foveation models depend on experiments still in training (§4.4, Appendix G). Each finding has a direct biological precedent which we discuss as convergent (not conclusive) evidence in §5.2.

2 Related Work

Cognitive maps in RL navigation agents Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

Foveated vision and gaze policies in deep learning Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

Probing neural representations Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.

What is new here Three things distinguish this work. (i) A *controlled* five-condition ablation with everything but the visual pathway held identical, including the coarse condition that decouples “no vision” from “vision the encoder cannot resolve”. (ii) A protocol-corrected probing methodology (deterministic action selection at probe time; stochastic sampling inflates probe R^2 via a 4-step quasi-static-trajectory artefact, Appendix A). (iii) Convergent behavioural validation via memory transplant and shortcut discovery, outside the linear-probe framework. Biological motivation and the parallel to animal-navigation research are integrated in §5.2 rather than recapitulated here.

3 Methods

3.1 Setup

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes; evaluation uses 18 held-out MP3D test scenes (1008 episodes). All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar), each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions train to $\sim 250\text{M}$ environment frames; the blind baseline trains to 342M to allow slower convergence from proprioception alone.²

The five visual conditions (Figure 1): *blind* (no visual input); *uniform* (full 256×256 RGB encoded by ResNet-18); *foveated (fix)* (eccentricity-dependent Gaussian blur, $\sigma_{\max} = 8$ quadratic falloff, gaze locked at image centre, same ResNet-18 encoder)³; *coarse* (48×48 uniform RGB; at this resolution ResNet-18’s 32-fold spatial downsampling collapses the feature map to 1×1 , removing spatial structure but preserving channel information); *foveated (learned)* (same blur model, gaze centre predicted by a lightweight MLP trained end-to-end). Coarse is the spatial-bandwidth lower bound for our encoder; it has visual input but its encoder output cannot resolve world-frame position. The 48×48 input also has lower spatial resolution than uniform’s 256×256 , so this single condition cannot fully separate the 1×1 collapse from low input resolution; the encoder-resolution scaling sweep (Appendix H) addresses this directly. The foveated conditions disable the running-mean-and-variance input normaliser to avoid an in-place autograd conflict along the gaze-decoder gradient path; an invariance control is in training. We also discovered and patched a silent NaN-gradient corruption mode in DD-PPO (Appendix F).

3.2 Probes

After training, all weights are frozen. We collect per-step top-layer LSTM hidden states (512-d) paired with ground-truth labels from $n=500$ Gibson held-out episodes under *deterministic* argmax-action rollouts (matching our shortcut / transplant / standard eval protocols; the alternative stochastic protocol gives trivial ~ 4 -step rollouts and is documented in Appendix A).⁴ We train Ridge probes ($\alpha = 10$) with episode-level 5-fold cross-validation, report mean $\pm \sigma$ across folds, and verify probe specificity with Hewitt–Liang label-permutation selectivity [Hewitt and Liang, 2019]. Probed variables: GPS position, compass heading, distance-to-goal, lag- k path history ($k \in \{0, \dots, 20\}$), visited-region occupancy, full occupancy grid, per-unit spatial information, ego-frame goal vector. For H2 representational geometry we additionally use linear CKA [Kornblith et al., 2019] and cross-condition probe transfer.

We complement probes with two interventions on the LSTM state. *Memory transplant*: paste a donor’s hidden state into the recipient at step 30 and compare baseline / self-transplant / cross-transplant SPL across 150 episodes per pair.⁵ *Shortcut discovery*: run paired same-scene-different-goal episodes (20

²Algorithm-and-task scope: DD-PPO + PointGoal in Habitat with ResNet-18 visual encoder. The H1/H2 patterns we report could in principle shift under different RL algorithms (e.g., on-policy A2C, IMPALA, off-policy SAC), different navigation tasks (ObjectGoal, ImageGoal, exploration), different simulators (iGibson, AI2-THOR, real-world), or different visual encoders (smaller CNN, ViT). Each is a deliberate scope choice for direct comparability with [Wijmans et al., 2023]; verifying generality is open.

³Foveation-model choice: Gaussian blur with quadratic eccentricity falloff is a tractable approximation, not a faithful model of biological retinal sampling. Real foveation involves spatially non-uniform sampling (log-polar or multi-scale pyramid) which directly reduces the encoder’s spatial output dimensionality, whereas Gaussian blur preserves it. The F3 log-polar control in training (Appendix G) is a falsifiable test of whether the encoder spatial-output story explains foveation under a different operationalisation; if log-polar matches uniform on H1 (rather than the predicted intermediate value), the H1 mechanism story would need reframing to depend on something other than encoder spatial output.

⁴We probe top-layer hidden state $\mathbf{h}_2$ in the main results because it is what the (linear) policy head reads. The cell state $\mathbf{c}_t$, lower-layer hidden states ($\mathbf{h}_0, \mathbf{h}_1$), and pre-LSTM concat vector are not probed in main figures (Figure 2c shows GPS at $\mathbf{h}_0/\mathbf{h}_1/\mathbf{h}_2$ as a single-fit per-condition layout but not the full per-condition CKA / transplant / shortcut machinery). If condition-specific structure lives in $\mathbf{c}_t$ or earlier layers but not $\mathbf{h}_2$, the H2 “disjoint subspaces” claim would extend or weaken accordingly.

⁵Transplant midpoint $t=30$ is a single design choice. The supplementary midpoint sweep across $\{0, 15, 30, 60, 90\}$ steps (Figure 8) shows the cross-self gap is roughly stable for $t \geq 15$, so the asymme-

scenes $\times$ 10 pairs) and compare second-episode SPL under reset vs. persistent LSTM. For foveated (fix), all analyses use ckpt.36 at $\sim 174\text{M}$ frames (last clean intermediate before the silent-corruption window).

4 Results

4.1 Five conditions, same task

Table 1 collects the headline per-condition numbers. *All five agents solve the same task at near-equivalent competence.* Sighted conditions reach 96–99% success; blind reaches 93% at a longer training budget, reproducing Wijmans et al. [Wijmans et al., 2023]. The narrow success bracket rules out a skill-gap account for any encoded-content differences below. The probe machinery is calibrated: distance-to-goal decodes everywhere ($R^2 \in [0.81, 0.90]$), demonstrating Ridge has the capacity to fit high- R^2 scalars when the hidden state contains them; Hewitt–Liang label-permutation selectivity is high precisely where GPS is high (Appendix A). The behavioural near-equivalence sets up the question the rest of §4 addresses: *what differs internally*, given that what the agents *do* is the same?

Table 1: Per-condition headline. Probe R^2 columns are the 5-fold episode-level CV mean $\pm$ std on full-length deterministic-rollout trajectories (no per-episode step cap). *GPS* and *Compass* are world-frame. The bottleneck-vs-rich-encoder gap is in temporal *stability* of the code, not its mid-episode existence (Figure 2b): all five conditions decode top-layer GPS at $R^2 \in [0.6, 0.95]$ in the typical-episode window (steps 50–200); the rich-encoder long-tail collapse is what produces the sub-chance numbers below. Bold marks the H1-relevant ordering (blind $>$ coarse $\gg$ rich-encoder).

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	342M	0.59	0.93	$+0.95 \pm 0.02$	$+0.81 \pm 0.08$
Coarse	250M	0.67	0.98	$+0.78 \pm 0.10$	$+0.64 \pm 0.10$
Uniform	250M	0.85	0.99	-0.31 ± 0.86	$+0.36 \pm 0.23$
Foveated (fix)	174M	0.78	0.96	$+0.06 \pm 0.88$	$+0.07 \pm 0.69$
Foveated (learned)	250M	0.82	0.98	-2.43 ± 3.98	-1.34 ± 3.14

4.2 Encoder–memory race: a temporal-stability claim (H1)

The headline finding is non-obvious: visual-encoder capacity tracks top-layer LSTM spatial encoding *inversely*. Conditions with the smallest visual-feature throughput per step (blind, coarse) carry a strong linear GPS code at the policy readout, and conditions with the richest visual stream (uniform, foveated, foveated (learned)) do not (Figure 2a). Goal-distance, included as a sanity check, decodes everywhere.⁶

Temporal stability is the precise claim Step-binned probing (Figure 2b) separates “never had a top-layer GPS code” from “had one mid-episode then lost linear readability”; rich-encoder conditions are the latter. The Table 1 “rich-encoder $\approx$ chance” headline is therefore long-tail behaviour, not mid-episode absence. *H1 is most precisely a temporal-stability claim about the linearly-decodable top-layer GPS code.* A lag- k probe corroborates: blind sustains GPS at $k = 20$, while rich-encoder conditions are at chance for every $k \leq 10$.

Layer specificity and non-linear probe. All five conditions have GPS at LSTM Layer 0 because the GPS-sensor embedding enters there (Figure 2c). The H1 distinction is what happens through Layers 1–2: bottleneck retain to Layer 2, rich-encoder discard. Since the DD-PPO action head is a linear projection from $\mathbf{h}_2$, this maps to “policy-accessible”. A 2-layer MLP on the same top-layer states recovers GPS for rich-encoder conditions ($R^2 \in [0.51, 0.73]$); H1 is therefore precisely about *linear* top-layer decodability, not whether *any* probe can recover position. The MLP result alone cannot

try pattern we report is not specific to $t=30$; later midpoints (e.g. $t \geq 100$) are not yet measured, and earlier midpoints ($t < 15$) collapse to a protocol-noise floor.

⁶Whether the correlation reflects causation is what the §4.4 experiments and the Appendix H resolution sweep test directly. All numerical claims in §4.2 are single-seed; multi-seed replication of all five conditions is in flight.

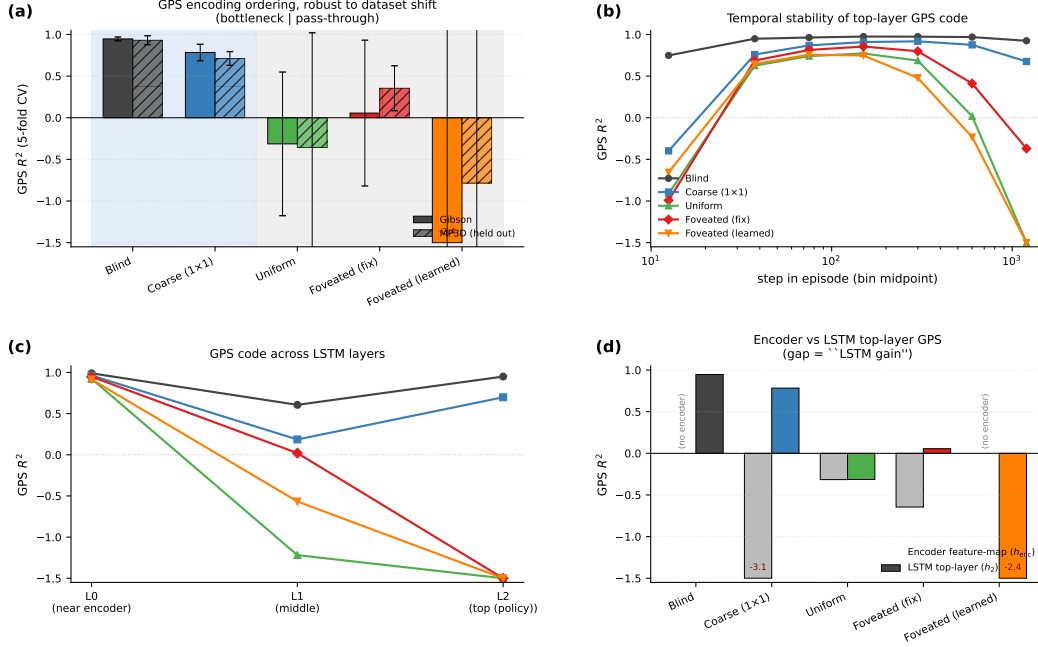


Figure 2: H1 evidence in four panels. (a) Top-layer GPS R^2 (5-fold CV, deterministic rollouts) on Gibson and held-out MP3D (300 episodes/condition). Blind and coarse (bottleneck) decode world-frame GPS on both datasets; uniform / foveated / foveated (learned) (rich-encoder) do not, except foveated recovering on MP3D. (b) Top-layer GPS R^2 binned by step-in-episode: all five conditions encode top-layer GPS in the typical window (steps ~ 25 –200); the code becomes no longer linearly readable in the rich-encoder long-tail (negative R^2 in the 800+ step bin), while bottleneck conditions hold $R^2 > 0.7$ to step 800+. The headline in (a) is therefore primarily long-tail behaviour. (c) Per-LSTM-layer GPS R^2 : all five conditions have GPS at Layer 0 (the GPS sensor embedding enters the LSTM input vector here — a sensor pass-through, not the network “computing” position); divergence is at Layer 2, the policy readout. (d) Encoder feature-map probe (grey bars; post-ResNet-18, pre-LSTM) vs LSTM top-layer probe (coloured bars). Blind and foveated (learned) have no encoder probe (no encoder / no encoder-probe data resp.). The grey bars sit near or below zero everywhere, so *no sighted encoder linearly carries GPS*; the coloured-grey gap is the “LSTM gain”. Coarse gains ~ 3.9 , uniform ~ 0 , foveated ~ 0.7 .

exclude exploitation of position-correlated features (wall distance, scene identifiers); Appendix B discusses both. The bottleneck-vs-pass-through partition is preserved on held-out MP3D (Figure 2a).⁷

Proposed mechanism The encoder feature-map probe (§4.4, Figure 2d) shows that none of the sighted encoders linearly re-derives GPS from its visual stream. The H1 phenomenon is therefore not encoder-level position retention; **we propose it is a substitution: the LSTM treats integrating its Layer-0 GPS-sensor input across time and exploiting the encoder’s position-correlated visual features as alternative routes to the same policy output**. Bottleneck conditions force the integration route (no rich visual features available), so the temporally-stable GPS code persists; rich-encoder conditions provide a viable visual route, so the integrated code is no longer needed by the policy and ceases to be linearly readable. The early-then-decay temporal profile in Figure 2b is the direct fingerprint this account predicts.⁸

⁷Two single-seed cross-dataset shifts on MP3D — foveated GPS at chance $\rightarrow +0.35$, foveated (learned) compass $-1.34 \rightarrow +0.41$ — await multi-seed replication; if they survive, they would suggest condition-specific MP3D recovery worth investigating, but at $N = 1$ we cannot rule out per-seed noise.

⁸A direct causal test — ablating the visual stream mid-rollout in rich-encoder agents to see whether the top-layer GPS code re-emerges, or perturbing the GPS sensor input mid-rollout in bottleneck agents to see whether SPL collapses — is left for future work.

Relation to Wijmans et al. [Wijmans et al., 2023]. They showed blind agents develop spatial representations; our blind result ($R^2 = 0.95$) replicates this and coarse extends it to a condition that has visual input but 1×1 spatial encoder output. The shared trigger is therefore minimal visual-feature variety per step, not absence of vision. The parallel to animal-navigation findings is treated separately as convergent (not mechanistic) evidence in §5.2.

4.3 Format-level divergence across conditions (H2)

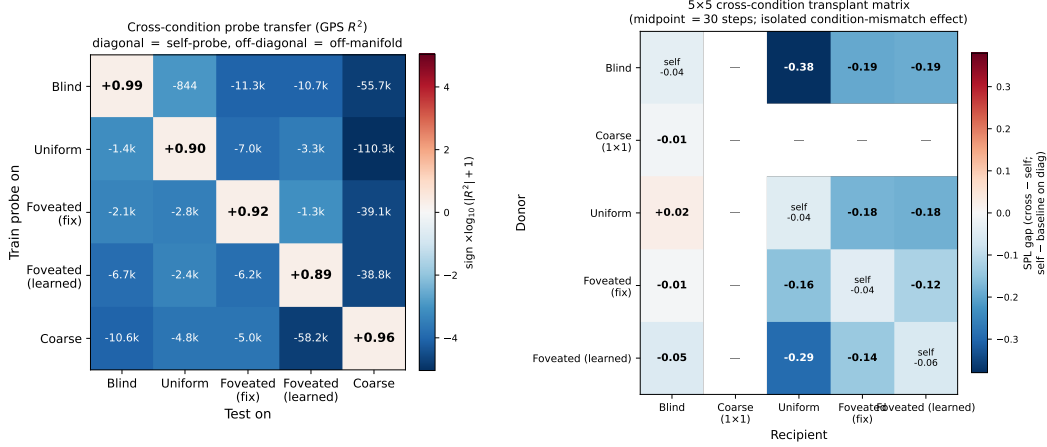


Figure 3: H2 convergent evidence, two views. *Left*: cross-condition probe transfer for GPS R^2 (train probe on row, test on column; signed-log colour scale). The diagonal is +0.89—+0.99 (self-probe), every off-diagonal predicts off-manifold ($R^2 \ll -800$, often $\ll -10^4$). *Right*: 5×5 cross-condition memory-transplant matrix at midpoint $t=30$. Cells off-diagonal: cross-transplant SPL minus recipient’s self-transplant SPL (isolated condition-mismatch effect). Cells with “—” (coarse as recipient): not yet measured. Both heatmaps are asymmetric: bottleneck-donor states hurt rich-encoder recipients far more than the reverse, and probes trained on bottleneck states fail catastrophically on rich-encoder hidden states. A midpoint-sweep companion across $\{0, 15, 30, 60, 90\}$ is in Appendix D.

Behavioural test (memory transplant) Pasting a donor’s mid-episode hidden state into a recipient’s policy at step 30 tests whether the representational geometry differences *matter* outside the probe framework (Figure 3, right). Three structural facts emerge that no linear-similarity measure on its own would surface. *Asymmetry along the bottleneck axis*: bottleneck-donor states are toxic to rich-encoder recipients while the reverse direction is benign — the rich-encoder LSTM cannot make sense of bottleneck-donor activity at the policy readout, but a bottleneck LSTM can still drive its policy from rich-encoder donor activity. *Recipient brittleness scales with encoder bandwidth*: uniform suffers the largest cost from *any* foreign donor; blind tolerates them all. *The linear-similarity blind spot*: pairs that read as indistinguishable under unaligned linear CKA differ by 0.01–0.38 SPL under transplant — “these representations look identical to a linear similarity metric” is consistent with “their actual policy-relevant content is incompatible”.⁹

Convergent evidence at the probe level. Cross-condition probe transfer (Figure 3, left) shows the same asymmetric pattern: probes trained on bottleneck states predict off-manifold by a wider margin on rich-encoder hidden states than the reverse. The 1-NN purity is 1.000 vs. chance 0.20 on 1500×5 pooled top-layer states — a sample’s nearest neighbour by Euclidean distance is always within its own condition.

Ruling out alternatives. Probe capacity is not the bottleneck (self-probes succeed everywhere); sample-size asymmetry is ruled out by common-sample truncation; task competence is bunched (96–99%), so the divergence is not a skill difference; probe transfer fails symmetrically, ruling out one condition’s code simply embedding another’s.

⁹The coarse-recipient column is currently unmeasured (4 cells queued); the recipient-ranking statement uses non-coarse-recipient cells only. Single-seed; the asymmetry pattern is robust to landed cells but the magnitudes will move with replication.

Boundaries. Our similarity tests are linear or near-linear; non-linear measures could reveal higher-order shared structure. Our five sensors are diverse but not exhaustive; hybrid sensors (sharp fovea + low-acuity periphery, depth-only) might produce intermediate geometries.

4.4 Foveation: convergence on H1, divergence on H2 + behaviour

Foveated (fix) and uniform fall in the same rich-encoder pass-through regime on the headline H1 metric (top-layer GPS R^2 at chance), but they are not interchangeable on H2 / behavioural / dataset-shift axes (Table 2). The take-away: *within the rich-encoder regime defined by H1, foveation and uniform produce internally distinct representations whose differences only the H2-and-beyond machinery makes visible.*

Table 2: Foveation vs. uniform at $\sigma_{\max} = 8$ Gaussian blur with the same ResNet-18 encoder. Same on H1, different on H2 / behavioural / generalisation. Sources: Table 1, Figure 3, Figure 4, Figure 2d. Data are deterministic-rollout single-seed.

Measure	Foveated (fix)	Uniform	Verdict
LSTM GPS R^2 (Gibson)	$+0.06 \pm 0.88$	-0.31 ± 0.86	both chance, <i>indistinguishable</i>
Encoder $\rightarrow$ GPS R^2	-0.65 ± 0.28	-0.32 ± 0.08	both negative, overlapping
LSTM compass R^2 (Gibson)	$+0.07 \pm 0.69$	$+0.36 \pm 0.23$	uniform $\sim 5\times$ better
LSTM GPS R^2 (MP3D, held out)	$+0.35 \pm 0.27$	-0.36 ± 1.38	differ by 0.71
Shortcut SPL drop %	21%	41%	uniform $2\times$ more brittle
Memory-transplant cost (fov $\rightarrow$ uni)	-0.10 SPL beyond self-noise		not interchangeable
Pairwise CKA(fov, uni)	$< 10^{-4}$		near-orthogonal subspaces
1-NN purity (pooled)	1.000 vs. chance 0.20		linearly separable

Two takeaways follow. *The H1 phenomenon lives inside the LSTM, not at the encoder.* Probing the encoder feature map directly shows that no sighted encoder linearly decodes GPS (Figure 2d, grey bars), so the bottleneck-vs-rich-encoder gap is not encoder-level position retention. *The LSTM-vs-encoder gap* (coarse gains ~ 3.9 , foveated ~ 0.7 , uniform ~ 0) orders conditions by how much the LSTM contributes *above* what the encoder already supplies; we interpret this as a gradient in how much the LSTM relies on visual features rather than integrated GPS-sensor signal as encoder spatial-feature variety grows. *Foveation is not a copy of uniform.* Even where they converge on the H1 metric, the encoder feature-map probe rules out encoder-level differences — so *whatever foveation does differently from uniform happens inside the LSTM in subspaces a linear-GPS probe is blind to.* The four in-flight foveation experiments (Appendix G) test *what* that something is; F3 log-polar in particular makes a falsifiable prediction: a $\sim 2\times 2$ encoder spatial output should produce LSTM GPS $R^2 \geq 0.3$, between coarse’s $+0.78$ and uniform’s ≈ 0 . If instead it matches uniform, the encoder-spatial-output mechanism would need reframing.¹⁰

4.5 Probe-readable vs. policy-used memory: a 2×2 dissociation

H1 measures what is *readable* from the recurrent state; the shortcut-discovery experiment measures what the policy actually *relies on*. Paired same-scene-different-goal episodes ($20 \text{ scenes} \times 10 \text{ pairs}$) compare second-episode SPL under reset vs. persistent LSTM: the SPL drop indexes how tightly the policy couples actions to integrated recurrent memory. Plotting it against probe-readable GPS (Figure 4) reveals two off-diagonal points: coarse has a linear top-layer GPS code that its policy weights below what the probe predicts, and uniform has no linear GPS *yet relies heavily on memory* — *presumably via non-spatial scene-history features the linear-GPS probe is blind to.* We label these candidate dissociations rather than established anomalies.¹¹

A trajectory-level look at *how* persistent-memory failures manifest is consistent with the dissociation (Figure 5). Aggregating across all same-floor paired-episode failures (Figure 5, right panel): uniform is the only condition with a meaningful sample size whose persistent agent ends up closer to the

¹⁰This interpretation is consistent with the data but not directly tested by a causal intervention; the resolution scaling sweep (Appendix H) tests it. All “LSTM gain” numbers and the foveation-vs-uniform contrasts in Table 2 are single-seed.

¹¹The 5 points are single-seed; confirming that “probe-readable” and “policy-relied-on” truly dissociate (rather than the off-diagonal points sitting within seed-to-seed noise) requires the multi-seed replication in flight.

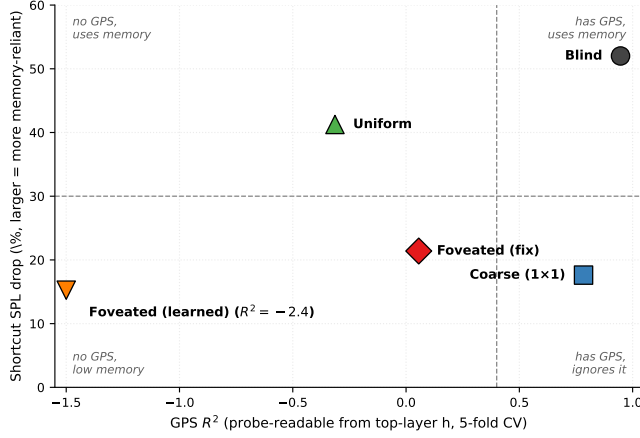


Figure 4: Probe-readable GPS (H1) vs. behavioural memory reliance (shortcut). The five conditions occupy four corners of a 2×2 : blind “has GPS, uses memory” (consistent with H1); foveated (fix) and foveated (learned) “no linear GPS, low memory reliance” (also consistent with H1); **coarse** and **uniform** are the two off-diagonal candidate dissociations (single-seed; multi-seed gates the “anomaly” label). Foveated (learned)’s $R^2 = -2.43$ is clipped at -1.5 .

previous-episode goal than the new one (margin $+1.83m$ on $n=46$); blind / coarse / foveated all end nearer the new goal but never reach it.¹² If the uniform pattern survives multi-seed replication, it would be consistent with the 2×2 : uniform’s memory would plausibly act as a visual / scene-history anchor that locks the agent in place, whereas blind’s GPS code would interfere via a different channel — the recurrent state mis-reporting current position to the policy’s goal-direction integrator.

4.6 Gaze location (H3, in progress)

H3 asks whether gaze *location* acts as a second representational-content axis within the rich-encoder regime. Our minimal learned-gaze module (deterministic sigmoid MLP, slow updates per 128-step segment) freezes at $(0.49, 0.62)$ with per-dimension $\text{std} \leq 2.5 \times 10^{-4}$ after 250M frames; anti-collapse regularisers restore variance only at task-cost (Appendix E). The collapse is not evidence about active gaze in general — the decoder is minimal by design — but it provides an empirical anchor for the gaze-location test: a foveated-shifted control with gaze hardcoded at $(0.49, 0.62)$ instead of $(0.5, 0.5)$, identical to foveated (fix) in every other respect, isolates static gaze location from the optimisation-path interaction that produced the collapsed gaze. Training is in progress at submission.

4.7 Boundaries

Three probes that do *not* discriminate define the boundary of our claims. Low-resolution (visited-region) occupancy decodes above chance everywhere; metric 20×20 occupancy fails everywhere (consistent with [Ramakrishnan et al., 2025] that high-resolution metric layout needs non-linear probes); the ego-frame goal vector is at chance everywhere because PointGoal supplies it as a per-step input. Population coding (Appendix D) sharpens H1 in a direction we did not expect: rich-encoder conditions have a few units with *higher* peak spatial information than blind, but those peaked units do not decode GPS. This suggests rich-encoder peaked units encode features correlated with position (visual landmarks, trajectory phase) rather than position itself; blind’s broader, lower-peak distribution carries the linearly-decodable position code.

¹²Foveated (learned) shares uniform’s positive-margin direction at $+2.30m$ but $n=5$ is too small to interpret. The locking interpretation is single-seed and not yet established; multi-seed replication is in flight.

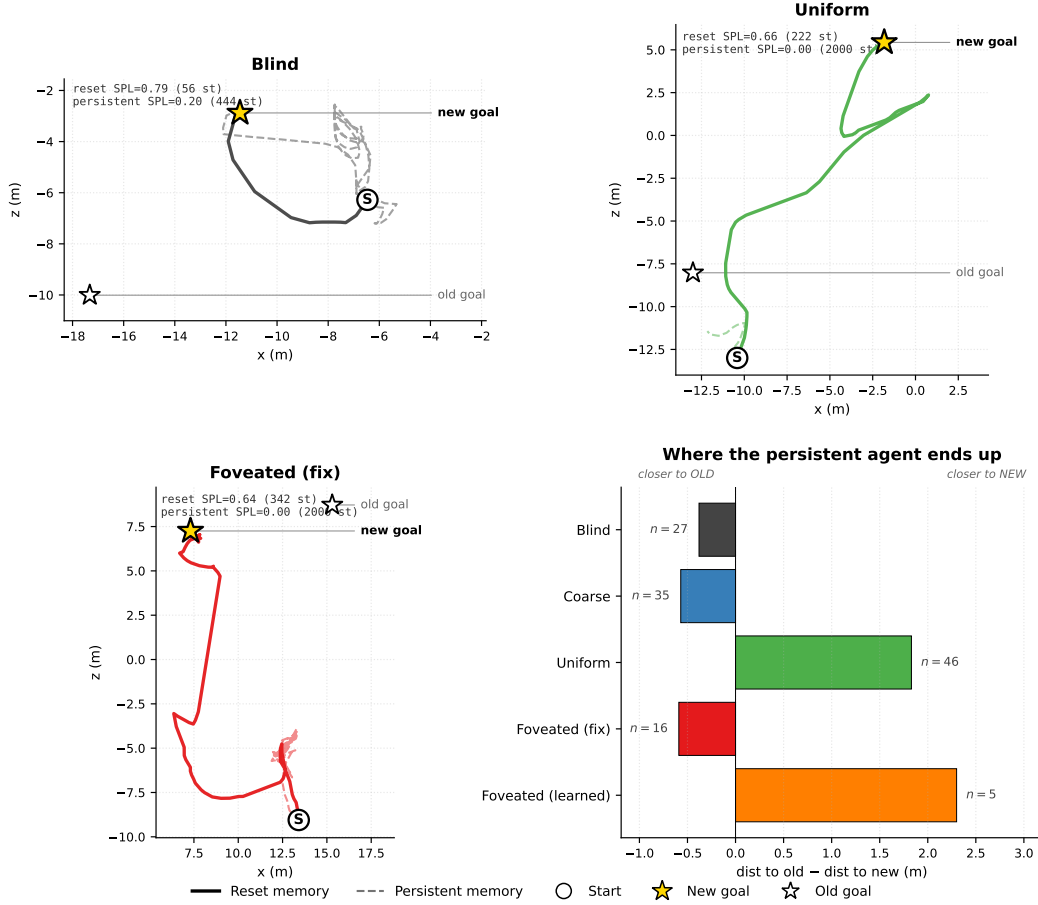


Figure 5: *Persistent-memory failure modes (one paired-episode case per condition + aggregate)*. The first three panels show *Blind* (persistent agent attempts the new goal, loops, gives up at 443 steps), *Uniform* (persistent agent stays at the old-goal location for the full 2000-step cap), and *Foveated* (persistent agent loops chaotically, reaches neither goal). Right panel: per-condition margin = (avg distance to old goal) – (avg distance to new goal) over all same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30 , $|\Delta y_{\text{goal}}| < 0.5\text{m}$). Uniform shows a positive margin at meaningful $n=46$ (closer to the old goal); foveated (learned) shows the same direction at +2.30m but $n=5$ is too small to interpret. Single-seed; multi-seed replication is in flight.

5 Discussion

5.1 Two orthogonal axes: sensor structure and gaze location

The three findings cohere along two axes. *Magnitude axis (H1)*: how much of the policy’s spatial competence is carried by integrated recurrent memory varies inversely with what the encoder hands the LSTM per step. *Format axis (H2)*: even within the rich-encoder regime where the magnitude axis collapses to “not via linear top-layer GPS”, the recurrent state still carries condition-specific content that is not interchangeable across conditions — something is being learned per condition, but in linearly disjoint subspaces. The 2×2 dissociation in §4.5 is what the two axes look like together: coarse has the H1 magnitude (linear GPS code) but its policy under-weights it; uniform lacks the H1 magnitude but the H2 format axis still carries a memory the policy heavily uses.

Read this way, the encoder–memory race (H1) is a single dimension of a larger picture: what the recurrent state carries is shaped by what the encoder doesn’t, and the format of that carrying is condition-specific even when the magnitude saturates. We do not claim the underlying mechanism is the same in coarse (where the encoder is structurally bottlenecked) and uniform (where the encoder

is rich but happens to discard linear GPS); they sit at different points on both axes. H3 (§4.6) probes whether gaze location is a third axis within the rich-encoder regime, perpendicular to both magnitude and format; the foveated-shifted control is in training.¹³

5.2 Biological precedent: sensor structure shapes hippocampal representation

The principle that sensor structure constrains spatial-memory format is not one we propose; biological evidence for it predates ours by decades. Our contribution is to isolate the principle in a controlled silicon model where the encoder, memory, policy, training distribution and reward are held fixed and only the visual sensor varies. Three threads of animal-navigation research are particularly close to the three findings.

H1 (sensor bottleneck → memory takes over). Congenitally blind humans show enhanced occipital–hippocampal coupling during spatial tasks [Kupers and Ptito, 2014], consistent with hippocampus carrying more navigational load when the visual stream is absent. Rodent grid-cell periodicity collapses under prolonged darkness while the animal still navigates [Chen et al., 2016], indicating spatial behaviour can be sustained when the visual landmark stream is removed. Both are biological analogs of our blind / coarse regime, where the LSTM — deprived of rich visual variety per step — retains a linear top-layer GPS code that is no longer linearly readable in the rich-encoder conditions.

H2 (different sensors → different spatial codes). The primate vs. rodent contrast in hippocampal cell types — view-cells dominant in primates, place-cells dominant in rodents — has been attributed in part to the foveated-with-saccades vs. near-panoramic visual sampling regime each species uses [Rolls, 2024]. Cross-modality remapping in echolocating bats, where the same neurons recode under vision vs. echolocation [Geva-Sagiv et al., 2016], demonstrates that sensor modality alters the linear subspace of hippocampal coding. Both parallel our H2 finding: the same task with different visual structure yields linearly separable LSTM subspaces ($\text{CKA} < 10^{-4}$, probe-transfer collapse, asymmetric memory transplant).

H3 (gaze location modulates memory encoding). In primates, hippocampal theta is gated by saccades, and the reliability of this saccade-gated signal predicts memory encoding success [Jutras et al., 2013, Tas et al., 2016]. Where the eye points is therefore not memory-neutral. Whether the analogous prediction holds for an LSTM gaze module is what §4.6 is testing (foveated-shifted causal control in training).

We treat these as convergent (not conclusive) evidence. Convergence across systems — and across silicon vs. neural — is suggestive of a sensor–memory coupling principle more general than any single architecture, but it does not show that the mechanisms are the same. Our experiments isolate the principle in a controlled, single-architecture setting; biological work establishes that the principle exists in animals at all. The two are complementary; neither is decisive alone.

5.3 Why this, not alternative accounts

Four alternative explanations are ruled out: skill difference (success bunched 93–99%), probe capacity (DtG decodes everywhere), sample-size or stochastic-sampling artefacts (Appendix A), and linear-probe artefact (memory transplant reproduces the ordering outside the probe framework; MLP probe shows the divergence is specific to *linear* top-layer decodability, which is what the linear DD-PPO action head reads). Full rebuttals in Appendix C.

5.4 Implications and scope

The encoder–memory race is most fundamentally a claim about the *interface* between an upstream encoder and a downstream memory module — what the memory module carries depends on what the encoder doesn’t. Our LSTM-based PointNav agent is one tractable instantiation. **We expect the principle to apply, with adapted measurements, to any architecture that maintains a persistent state across timesteps and reads it via a linear or near-linear policy head:** transformer-based navigators (where “memory” lives in the KV cache or attention over history) should show analogous content-level dissociations when attended tokens are constrained the way our encoder bottleneck constrains

¹³4 coarse-recipient transplant cells are still pending; the asymmetry-direction along the magnitude axis is robust to landed cells, but completing the matrix is needed before the asymmetry-statement is exhaustive. The unifying account is a candidate; a direct causal test (visual ablation mid-rollout) is left for follow-up.

spatial features. The relevant axis is not pixel count: coarse’s 48^2 input with a 1×1 feature map has a strong linear top-layer GPS code while uniform’s 256^2 does not, so we attribute the pattern to encoder spatial-feature *variety* per step rather than raw input bandwidth. Within recurrent agents specifically, encoder capacity becomes a practical training-time lever for inducing cognitive-map-like representations *if* the resolution scaling sweep (Appendix H) confirms causality. Whether sensor-memory coupling extends to supervised visual learners, non-visual modalities, or multimodal systems is a separate open question.

5.5 Limitations

(i) *Single seed* for every cross-condition number reported in this paper (H1 R^2 values, LSTM gain numbers, 5×5 transplant cells, shortcut SPL drops, lock-onto-old margins, MP3D cross-dataset shifts); multi-seed replication is in flight and gates the strength of every quantitative claim. (ii) *Foveated (fix) checkpoint* is ckpt.36 (~ 174 M frames, last clean before a silent NaN-gradient corruption; Appendix F); the other four conditions use the final converged checkpoint. (iii) *Learned-gaze module* is a minimal deterministic sigmoid MLP that collapsed under pure task supervision; the H3 test (foveated-shifted control) does not depend on this module working. Richer gaze mechanisms (stochastic gaze, intrinsic-reward gaze, attention-based selection) are out of scope. (iv) *Foveation model*: $\sigma_{\max} = 8$ Gaussian blur is a simplification; a log-polar control and an $\sigma_{\max} \in \{2, 4, 12, 20\}$ strength sweep are in training (Appendix G). (v) *Linear probes only*: non-linear probes might reveal structure Ridge misses (a 2-layer MLP partially recovers GPS in rich-encoder conditions; Appendix B). (vi) 5×5 transplant matrix has 4 unmeasured coarse-recipient cells (Figure 3, right); the asymmetry statement uses non-coarse-recipient cells only. (vii) *Architecture scope*: we use a recurrent (LSTM) backbone as a single-architecture case study for direct comparability with the emergent-maps literature [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]. The encoder-memory race is a claim about an upstream-encoder $\rightarrow$ downstream-memory interface and should in principle apply to other architectures (transformer-based navigators, attention over history); we do not verify this here.

6 Conclusion

Using a recurrent (LSTM) PointNav agent as a controlled case study, a five-condition visual-pathway ablation suggests sensor structure as a representational-content axis: the encoder-memory race (H1), condition-specific representational format (H2), and a candidate probe-readable-vs.-policy-used dissociation at the H1 $\times$ shortcut intersection. The H1 ordering reproduces on held-out MP3D, and the pattern parallels independent animal-navigation findings on hippocampal compensation under sensory deprivation and cross-modality remapping. The cross-condition observations are single-seed; multi-seed replication is in flight and gates the strength of every quantitative claim. Pending experiments that would tighten or falsify the account: a finer-grained encoder-resolution scaling sweep (Appendix H) for causal H1; the foveated-shifted control for H3; the four coarse-recipient transplant cells; the foveation strength sweep and log-polar control. The encoder-memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators or other persistent-state architectures; we do not verify this here.

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A Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4 -step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

Length-matching ablation. Under deterministic rollouts, episode lengths span ~ 100 – 400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. **[TODO: A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect it to leave the H1 ordering unchanged because the bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.]**

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A , test on opposite heading). Our implementation returned “insufficient samples” sentinels at our probe sizes; not in our results.

B Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder’s flattened output, and feeds the result as the LSTM’s Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 2c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states + GPS labels as the linear Ridge probe, with the same 5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, foveated (learned) 0.73, uniform 0.55 (vs. Ridge’s chance R^2). The MLP probe alone cannot disentangle “non-linear position code” from “MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)”; the resolution scaling sweep (Appendix H) tests this more directly by varying input resolution at fixed encoder stack.

C Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all five conditions ($R^2 \geq 0.81$), demonstrating Ridge has the capacity to fit high- R^2 scalars when the hidden state contains them. Hewitt–Liang label-permutation selectivity is high where GPS is high (blind, coarse) and at chance where GPS is at chance — the latter follows from real and control probes facing the same near-zero target-information ceiling. The chance-level rich-encoder R^2 is therefore a hidden-state property, not a probe artefact.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with ~ 100 – 400 -step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix A.

Linear-probe artefacts. Two complementary checks. (a) *Behavioural*: the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-pass-through ordering; the 5×5 matrix’s asymmetry adds a structure no linear-similarity measure can read. (b) *Non-linear probe*: a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix B). The bottleneck-vs-pass-through divergence is therefore not about whether *any* probe can recover position; it is specifically about whether a *linear* top-layer probe can. Since the DD-PPO action head is a linear projection from $\mathbf{h}_2$, the linear metric is what the policy itself can use, so H1 maps onto policy-accessible encoding regardless of how the MLP interpretation resolves.

D Supplementary visualisations

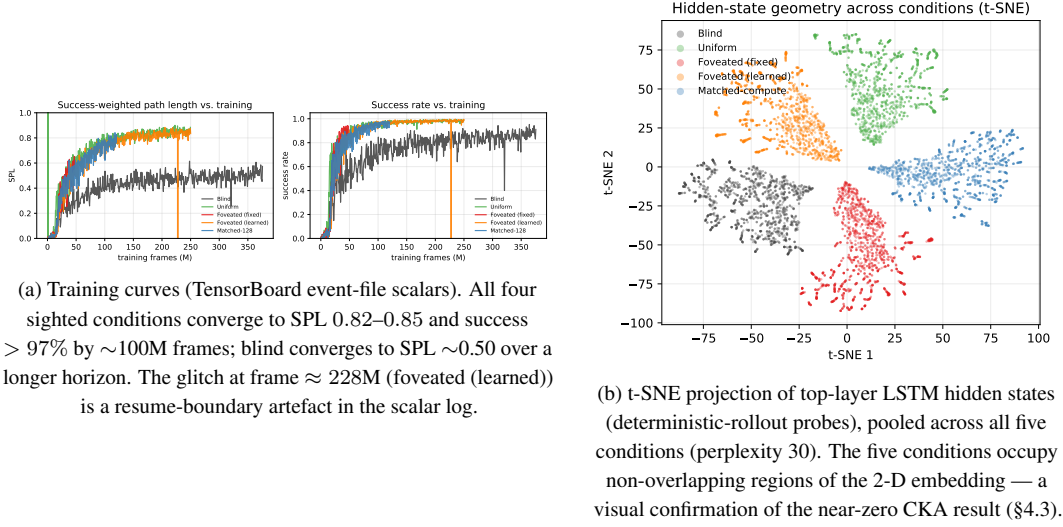


Figure 6: Supplementary visualisations referenced from the main text.

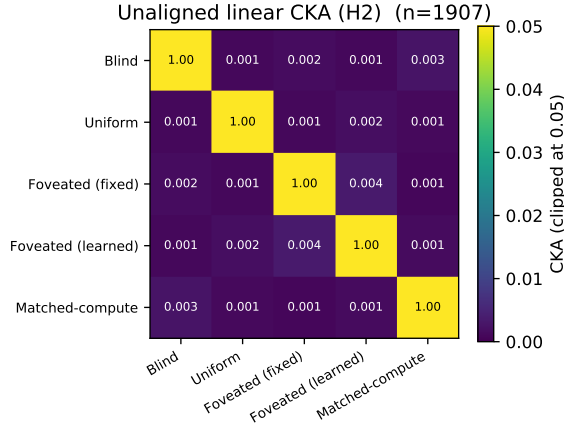


Figure 7: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states ($n=95,640$ steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4} . Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure 3); CKA is the least informative of the three H2 measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

E Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation. In the first foveated (learned) training run the gaze decoder collapsed (per-dimension $\text{std} \approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \rightarrow 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

Target-variance hinge. We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma_{\text{target}}^2 - \text{Var}_{\text{batch}}[g])$ where $g \in [0, 1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma_{\text{target}}^2 = 0.01$ ($\text{std} \approx 0.1$), $\lambda = 0.01$. The hinge disappears once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \text{Var}[g]$ variant as a negative control.

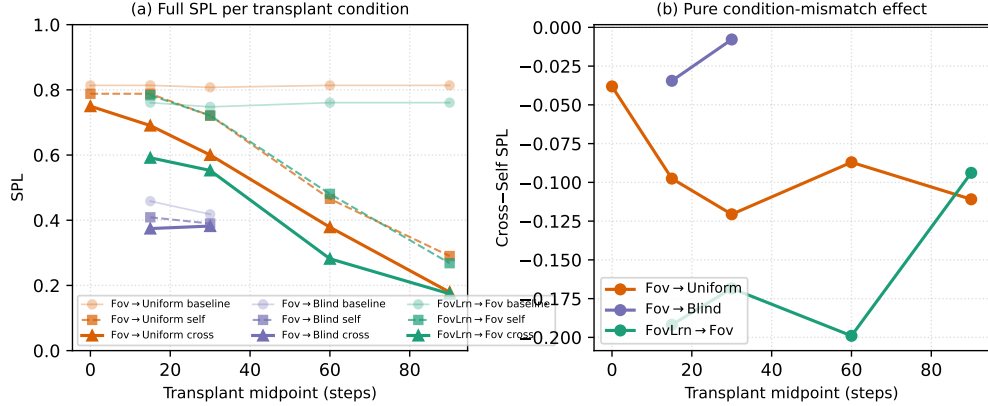


Figure 8: Memory-transplant midpoint sweep on three representative donor→recipient pairs (foveated→uniform, foveated→blind, foveated (learned)→foveated) across midpoints $\{0, 15, 30, 60, 90\}$ time-steps ($n=100\text{--}150$ episodes/cell). The full 5×5 matrix in Figure 3 (right) is at midpoint $t=30$; this supplementary figure shows that the cross-self gap is roughly stable for $t \geq 15$ and that the $t=0$ self-/cross- baseline collapses to a protocol-noise floor (so the diagonal in Figure 3 is correctly read as “rollout-divergence noise” rather than “condition-mismatch noise”). Foveated (learned)→foveated incurs -0.17 SPL on average across $t \geq 15$, foveated→uniform -0.10 , foveated→blind -0.02 , consistent with the asymmetry pattern in the full matrix.

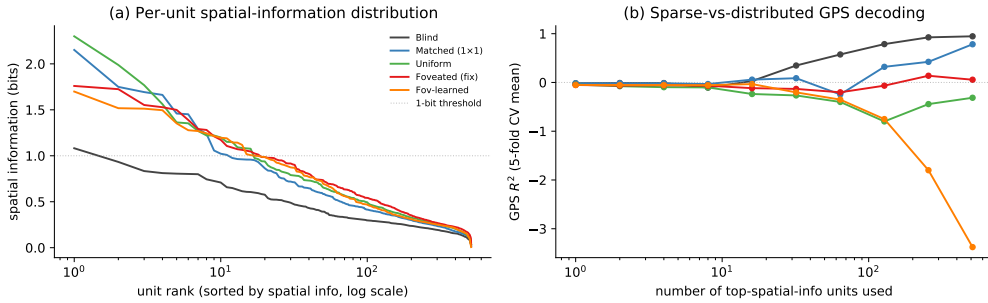


Figure 9: Population coding analysis (referenced from §4.7). (a) Per-unit spatial-information distribution: each curve is one condition’s units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform / foveated / foveated (learned)) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). (b) Sparse-vs-distributed GPS decoding: probe R^2 when only the top- k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information $\neq$ position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

Pilot 1 (uncapped, job 2840256). After 10 M frames, gaze variance exceeded that of a uniform distribution over $[0, 1]^2$ (std $[0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at ~ 8 M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288). Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with std $[0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

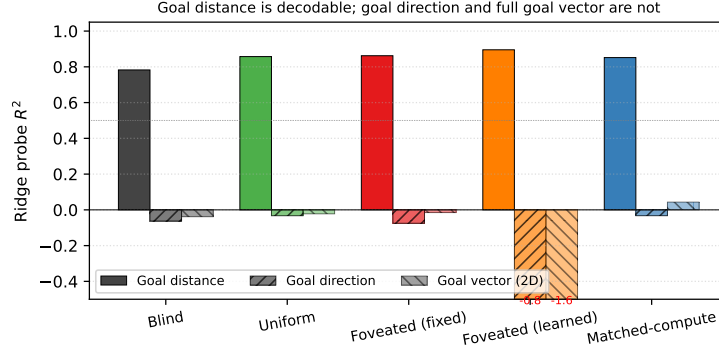


Figure 10: Goal-vector probe R^2 (referenced from §4.7). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$, validating probe machinery); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

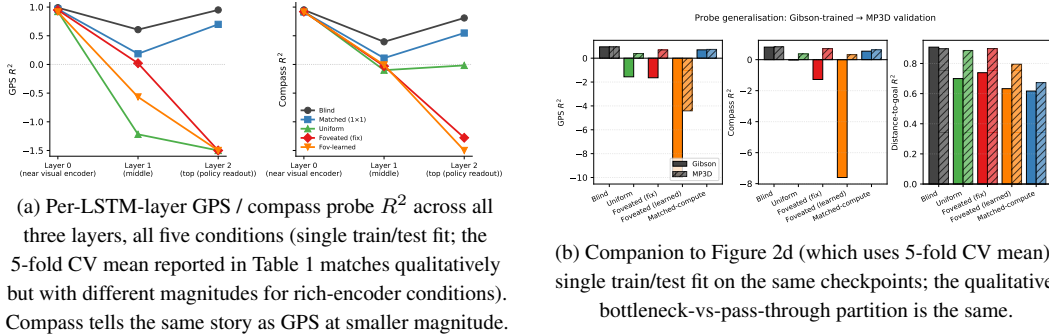


Figure 11: Per-layer and MP3D companion figures (referenced from §4.2).

Interpretation. Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze location with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

F Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈ 170 M frames of stable progress (reward ≈ 10 , SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *=` NaN remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL= 0 until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-

head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PP0.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.

G Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from §4.4. All four are submitted; numbers will populate §4.4 in revision as they land.

Table 3: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	in training
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training
Encoder feature-map probe (coarse / uniform / foveated)	where position information is held	landed; see §4.4
Foveated-shifted (gaze (0.49, 0.62))	gaze-location axis (links to H3)	in training

Foveation strength sweep (F1–F4). We train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve directly tests the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 8×8 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2 \times 2$, much closer to coarse’s 1×1 than to uniform’s 8×8 . This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*. Falsifiable prediction (written before result available): log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse’s $+0.78$ and uniform’s ≈ 0 , with corresponding behavioural shifts. If instead log-polar matches uniform, the encoder-spatial-output-dimensionality mechanism is wrong. F3 in training ($\sim 14\text{M} / 250\text{M}$ frames at submission, ETA 2–3 days).

Foveated-shifted (gaze location). Training a foveated agent with hardcoded static gaze at (0.49, 0.62) matches foveated (learned)’s collapsed gaze location while removing the learned-gaze module from the optimisation path. If foveated-shifted does *not* reproduce foveated (learned)’s specific behaviours (Gibson–MP3D compass swing, transplant ranking), those behaviours can be attributed to the optimisation-path interaction rather than to gaze location alone. Links to H3 (§4.6).

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists vs. what is in flight, and (ii) the reader can identify which incoming result would change the paper’s interpretation.

H Encoder-capacity scaling sweep

[TODO: The scaling sweep trains coarse variants at input resolutions $\{32, 48, 64, 96, 128, 192\}$ alongside the existing blind (no encoder) and uniform 256×256 anchors. Probe results on the trained checkpoints will populate Figure 12 and this appendix section. Hypothesis: GPS probe R^2 decreases monotonically with input resolution, from $R^2 \approx 0.95$ at no-encoder to $R^2 \approx 0$ at 256×256 , crossing into the pass-through regime between 48 and 128 pixels.]

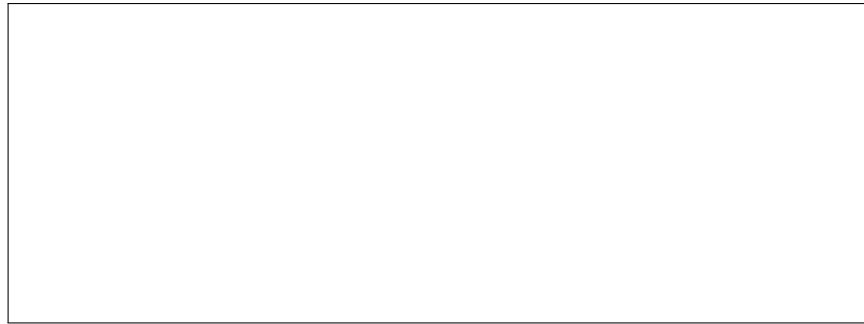


Figure 12: Encoder-capacity scaling curve. **[TODO: Pending data.]** x-axis: input resolution in pixels; y-axis: GPS and compass probe R^2 (5-fold CV, mean $\pm\sigma$).